VIGIL

An open-source reliability workspace



Aman Kumar
DevSecOps capstone
16 October 2026

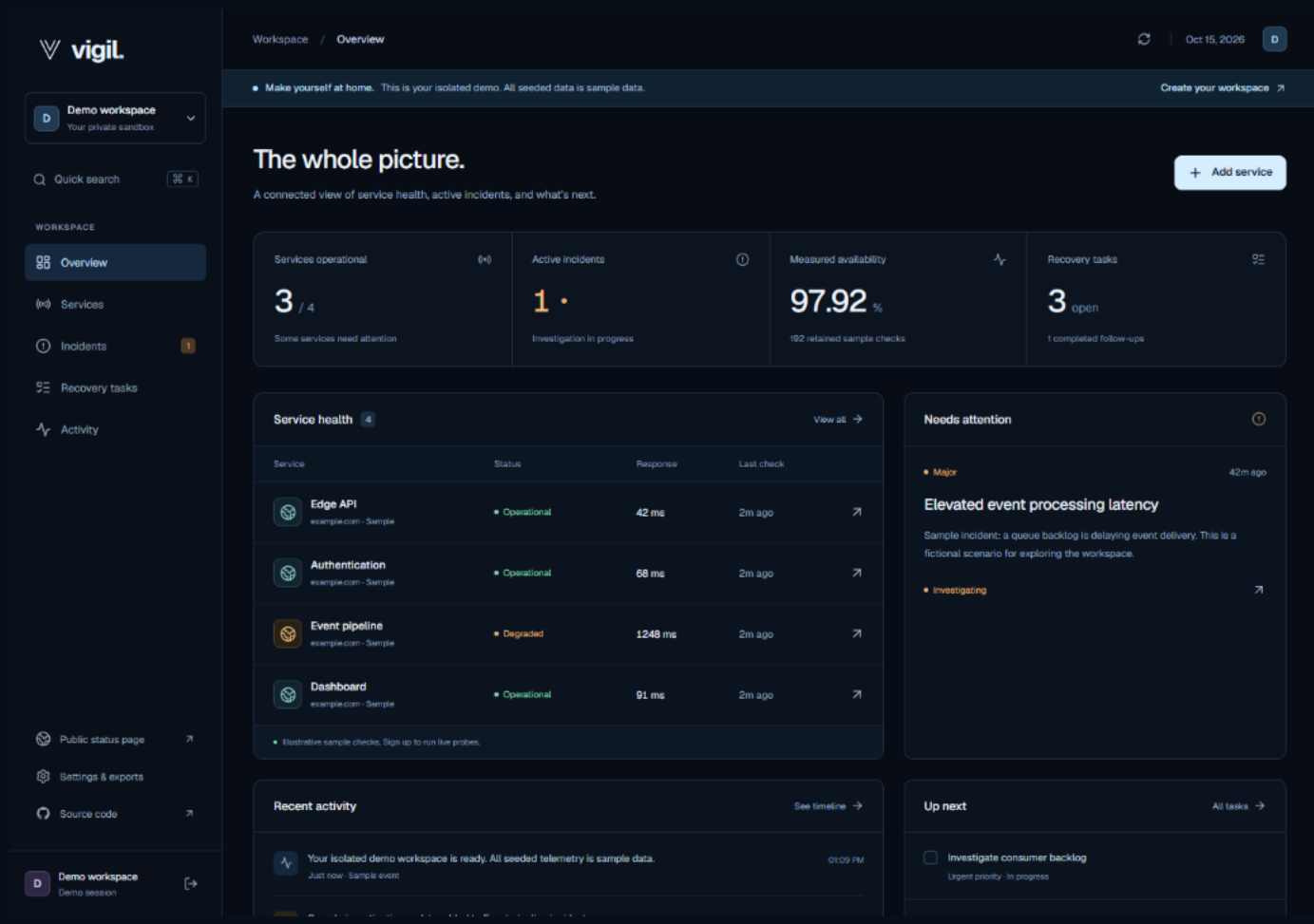
Service health and response

Monitor

Public HTTPS checks, status pages, and SVG badges.

Respond

Incident timelines, recovery tasks, and factual postmortems.



Demo telemetry is visibly labeled.

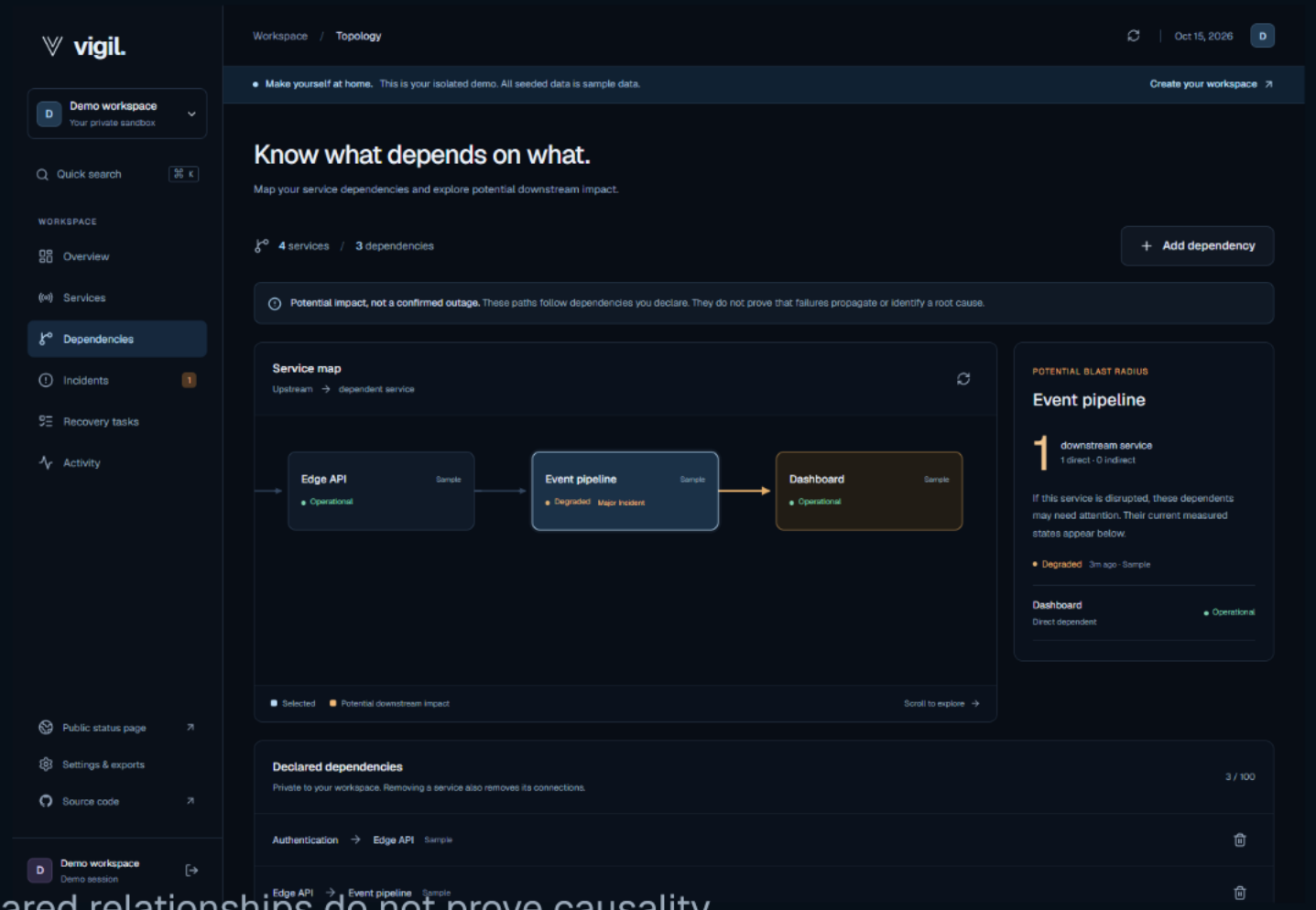
Service dependency analysis

Potential impact

Direct and indirect dependents

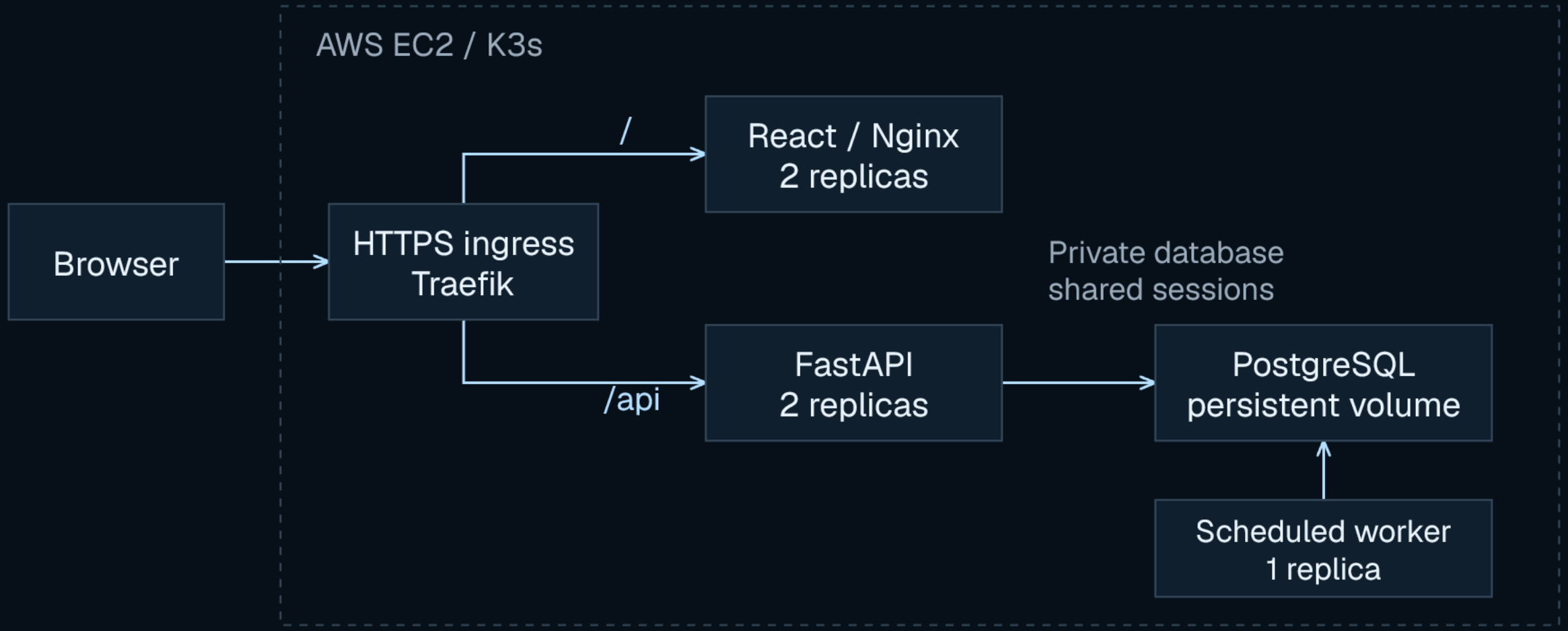
Current health and incident severity

Private, persistent connections



Declared relationships do not prove causality.

Application architecture



API and worker perform bounded, validated public HTTPS probes.

Implementation stack

Layer	Implementation	Purpose
API	Python / FastAPI	Validated, owner-scoped REST routes
Data	PostgreSQL / SQLAlchemy / Alembic	Persistent sessions and versioned schema
Frontend	React / TypeScript / Vite	Responsive app and original WebGL
Delivery	Docker / Actions / Trivy / GHCR	Tested and scanned SHA images
Runtime	Terraform / K3s / Helm	Reproducible AWS deployment

Automated verification

128

automated checks

Isolated test data
Real PostgreSQL smoke
Real Compose restart

Suite	Passed
API behavior	94
Frontend behavior	24
AWS spending guard	10

Verified CI revision dbe4044

CI and security gates



0

HIGH or CRITICAL findings
in both published runtime images

Unfixed findings included. No exclusions. Publication requires every gate.

AWS Free plan deployment

t3.small

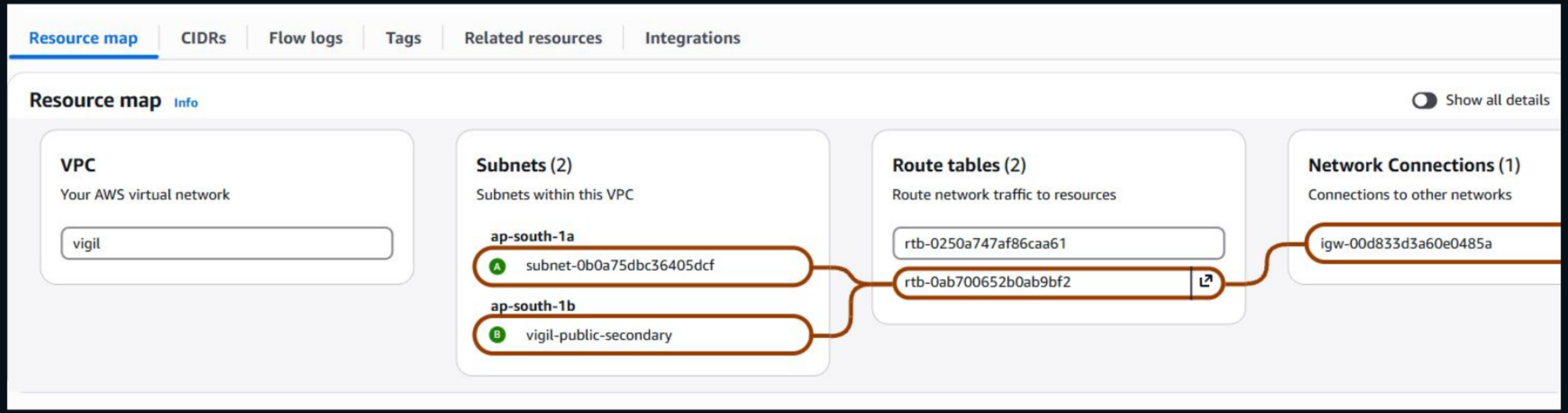
2 GiB / EC2 / K3s
Mumbai region

Account boundary

FREE and ACTIVE
Positive finite credits
No paid-plan upgrade

Rubric gap: EKS

Source validated.
Apply and destroy
remain unverified.



Kubernetes rollout and recovery

2 API 2 frontend 1 worker

Frontend

Recreate avoids mixed static asset versions.

API

Zero surge reduces peak rollout memory.

Recovery

Migrations gate startup. Route errors offer retry.

HPA: 2 to 4, then back to 2

Six readiness checks passed on the 2 GiB host.

GitOps with Argo CD



Synced and Healthy

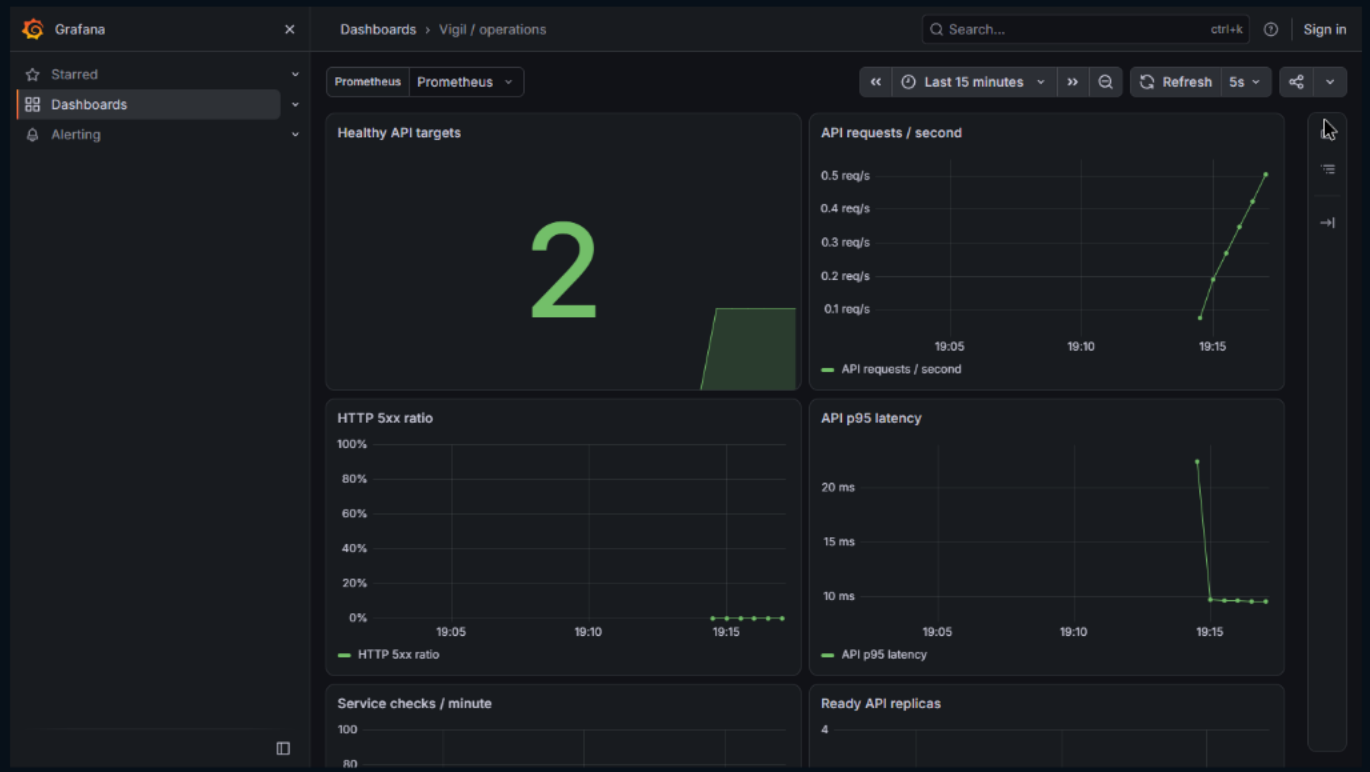
vigil-preview / automatic Git promotion
Synced Git revision 33f8e1f4e01e
Production remains owned by Helm.

Live observability

Actual application metrics

Healthy API targets
Request rate and p95 latency
HTTP 5xx ratio

Recorded AWS exercise.
Private monitoring stack.



Actual AWS capture / 16 October 2026.

Lessons and remaining limits

Small-host releases

Peak deployment memory and
asset consistency matter.

Known limits

Single host and finite credits
No team invitations or email recovery
EKS apply / destroy unverified

Evidence stays specific

Separate tests, source, deployment,
and observed behavior.

github.com/ReaperXD67/vigil